

MAT 310, Handout 18: Multivariate Distributions (5.1, 5.2)

Motivation: We have primarily focused on one RV at a time; however, there are times we may be interested in the values of 2 or more RVs simultaneously. For example, a medical researcher may be interested in the height and weight pairs of the patient population.

- Univariate Distribution:
- Bivariate Distribution:
- Multivariate Distribution:

Let Y_1 and Y_2 be discrete RVs. The **joint probability (mass) function** of $Y = (Y_1, Y_2)$ is given by

$$p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2)$$

for $-\infty < y_1 < \infty$, $-\infty < y_2 < \infty$.

Note: In the case of two discrete RVs (as above), the *joint probability (mass) function* may also be called the *bivariate probability (mass) function*.

Example 1. Suppose you roll 1 die and flip 1 coin. List the sample space, and then find the probability that you roll a 3 and flip a tail.

Properties of the Joint Probability (Mass) Function:

1. $p(y_1, y_2) \geq 0$ for all y_1, y_2
2. $\sum_{y_1} \sum_{y_2} p(y_1, y_2) = 1$

For any RVs Y_1 and Y_2 , the **joint (cumulative) distribution function**, $F(y_1, y_2)$, is

$$F(y_1, y_2) = P(Y_1 \leq y_1, Y_2 \leq y_2), \quad -\infty < y_1 < \infty, -\infty < y_2 < \infty.$$

Note: In the case of two RVs (as above), the *joint (cumulative) distribution function* may also be called the *bivariate (cumulative) distribution function*.

Example 2. Roll 2 dice. Let Y_1 be the # of odd numbers rolled and Y_2 be the sum of the 2 dice. Find $F(1, 5)$.

Properties of the Joint (Cumulative) Distribution Function:

1. $\lim_{(y_1, y_2) \rightarrow (-\infty, -\infty)} F(y_1, y_2) =$
2. $\lim_{(y_1, y_2) \rightarrow (\infty, \infty)} F(y_1, y_2) =$
3. $\lim_{y_1 \rightarrow -\infty} F(y_1, y_2) = \lim_{y_2 \rightarrow -\infty} F(y_1, y_2) =$

Let Y_1 and Y_2 be continuous RVs with joint cdf $F(y_1, y_2)$. If there exists a nonnegative function $f(y_1, y_2)$ such that

$$F(y_1, y_2) = \int_{-\infty}^{y_1} \int_{-\infty}^{y_2} f(t_1, t_2) dt_2 dt_1$$

for all $-\infty < y_1 < \infty$ and $-\infty < y_2 < \infty$, then Y_1 and Y_2 are said to be **jointly continuous**. The function $f(y_1, y_2)$ is called the **joint probability density function**.

Notes:

- Two RVs are jointly continuous if their joint cdf is continuous.
- In the case of two continuous RVs (as above), the *joint probability density function* may also be called the *bivariate probability density function*.

The double integral of the joint pdf allows us to calculate probabilities that Y_1 and Y_2 are in specified intervals:

$$P(a_1 \leq Y_1 \leq a_2, b_1 \leq Y_2 \leq b_2) = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(y_1, y_2) dy_2 dy_1.$$

Properties of the Joint Probability Density Function:

1. $f(y_1, y_2) \geq 0$ for all y_1, y_2
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y_1, y_2) dy_2 dy_1 = 1$

Example 3. Suppose Y_1 and Y_2 have the following joint pdf:

$$f(y_1, y_2) = \begin{cases} k, & 0 \leq y_1 \leq 2, 0 \leq y_2 \leq 3 \\ 0, & o.w. \end{cases}.$$

Note: because the joint pdf is constant, we say that $f(y_1, y_2)$ is *uniformly distributed* over the region $[0, 2] \times [0, 3]$.

(a) Find k .

(b) $P(Y_1 \leq 1, Y_2 \leq 2) =$

(c) $P(Y_1 \leq 2Y_2) =$

(d) $P(Y_1 + Y_2 \leq 1) =$

Example 4. Suppose Y_1 and Y_2 have the following joint pdf:

$$f(y_1, y_2) = \begin{cases} 3y_1, & 0 \leq y_2 \leq y_1 \leq 1 \\ 0, & o.w. \end{cases}.$$

Find $F(1/2, 1/3)$.